

Lab-3: MySQL Functions

String Functions

1. Upper and Lower

Code:

```
SELECT first_name, LOWER(first_name), UPPER(last_name) FROM employees;
```

Output:

+-----+			
Tables_in_sandbox_db			
+-----+			
employees			
+-----+			
+-----+-----+			
first_name LOWER(first_name) UPPER(last_name)			
+-----+-----+			
Arun	arun	KUMAR	
Priya	priya	SHARMA	
Rahul	rahul	VERMA	
Sneha	sneha	REDDY	
Vikram	vikram	PATEL	
Anjali	anjali	NAIR	
Karthik	karthik	IYER	
Meena	meena	JOSHI	
Suresh	suresh	MENON	
Divya	divya	KRISHNAN	
+-----+-----+			

2. Concat

Code:

```
SELECT CONCAT(first_name, ' ', last_name) AS full_name FROM employees;
```

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employees             |
+-----+

+-----+
| full_name             |
+-----+
| Arun Kumar           |
| Priya Sharma         |
| Rahul Verma          |
| Sneha Reddy          |
| Vikram Patel         |
| Anjali Nair          |
| Karthik Iyer         |
| Meena Joshi          |
| Suresh Menon         |
| Divya Krishnan       |
+-----+
```

3. Length

Code:

```
SELECT first_name, LENGTH(first_name) AS name_length FROM employees;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
employees	
+-----+	
+-----+-----+	
first_name	name_length
+-----+-----+	
Arun	4
Priya	5
Rahul	5
Sneha	5
Vikram	6
Anjali	6
Karthik	7
Meena	5
Suresh	6
Divya	5
+-----+	

4. Substring

Code:

```
SELECT first_name, SUBSTRING(first_name, 1, 3) AS short_name FROM employees;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
employees	
+-----+	
+-----+	
first_name short_name	
+-----+	
Arun	Aru
Priya	Pri
Rahul	Rah
Sneha	Sne
Vikram	Vik
Anjali	Anj
Karthik	Kar
Meena	Mee
Suresh	Sur
Divya	Div
+-----+	

Numeric Functions

1. Round

Code:

```
SELECT salary, ROUND(salary) AS rounded_salary FROM employees;
```

Output:

+-----+-----+	
Tables_in_sandbox_db	
+-----+-----+	
employees	
+-----+-----+	
+-----+-----+	
salary	rounded_salary
+-----+-----+	
55000.50	55001
48000.75	48001
62000.25	62000
71000.90	71001
45000.40	45000
52000.60	52001
68000.35	68000
59000.80	59001
47000.20	47000
51000.55	51001
+-----+-----+	

2. Ceil/Ceiling

Code:

```
SELECT salary, CEIL(salary) AS ceiling_salary FROM employees;
```

Output:

Tables_in_sandbox_db	
employees	
salary	ceiling_salary
55000.50	55001
48000.75	48001
62000.25	62001
71000.90	71001
45000.40	45001
52000.60	52001
68000.35	68001
59000.80	59001
47000.20	47001
51000.55	51001

3. Floor

Code:

```
SELECT salary, FLOOR(salary) AS floor_salary FROM employees;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
employees	
+-----+	
+-----+-----+	
salary	floor_salary
+-----+-----+	
55000.50	55000
48000.75	48000
62000.25	62000
71000.90	71000
45000.40	45000
52000.60	52000
68000.35	68000
59000.80	59000
47000.20	47000
51000.55	51000
+-----+-----+	

4. ABS

Code:

```
SELECT ABS(-250) AS result;
```

Output:

Output:

```
+-----+
| Tables_in_sandbox_db |
+-----+
| employees             |
+-----+
+-----+
| result |
+-----+
|      250 |
+-----+
```

5. Mod

Code:

```
SELECT emp_id, MOD(emp_id, 2) AS remainder FROM employees;
```

Output:

+-----+	
Tables_in_sandbox_db	
+-----+	
employees	
+-----+	
+-----+	
emp_id remainder	
+-----+	
101	1
102	0
103	1
104	0
105	1
106	0
107	1
108	0
109	1
110	0
+-----+	

Date Functions

1. Year

Code:

```
SELECT first_name, hire_date, YEAR(hire_date) AS hire_year  
FROM employees;
```

Output:

+-----+		
Tables_in_sandbox_db		
+-----+		
employees		
+-----+		
+-----+-----+		
first_name	hire_date	hire_year
+-----+-----+		
Arun	2020-06-15	2020
Priya	2021-02-10	2021
Rahul	2019-11-25	2019
Sneha	2018-08-30	2018
Vikram	2022-04-18	2022
Anjali	2020-10-12	2020
Karthik	2017-01-20	2017
Meena	2021-09-05	2021
Suresh	2023-03-15	2023
Divya	2022-12-01	2022
+-----+-----+		

2. Month

Code:

```
SELECT first_name, hire_date, MONTH(hire_date) AS hire_month  
FROM employees;
```

Output:

+-----+		
Tables_in_sandbox_db		
+-----+		
employees		
+-----+		
+-----+-----+		
first_name	hire_date	hire_month
+-----+-----+		
Arun	2020-06-15	6
Priya	2021-02-10	2
Rahul	2019-11-25	11
Sneha	2018-08-30	8
Vikram	2022-04-18	4
Anjali	2020-10-12	10
Karthik	2017-01-20	1
Meena	2021-09-05	9
Suresh	2023-03-15	3
Divya	2022-12-01	12
+-----+-----+		

3. Day

Code:

```
SELECT first_name, hire_date, DAY(hire_date) AS hire_day  
FROM employees;
```

Output:

Tables_in_sandbox_db		
employees		
first_name	hire_date	hire_day
Arun	2020-06-15	15
Priya	2021-02-10	10
Rahul	2019-11-25	25
Sneha	2018-08-30	30
Vikram	2022-04-18	18
Anjali	2020-10-12	12
Karthik	2017-01-20	20
Meena	2021-09-05	5
Suresh	2023-03-15	15
Divya	2022-12-01	1

4. TimeStampDiff

Code:

```
SELECT first_name,birth_date,  
TIMESTAMPDIFF(YEAR, birth_date, CURDATE()) AS age  
FROM employees;
```

Output:

+-----+			
Tables_in_sandbox_db			
+-----+			
employees			
+-----+			
+-----+-----+			
first_name birth_date age			
+-----+-----+			
Arun	1995-03-20	31	
Priya	1997-07-12	29	
Rahul	1993-01-05	33	
Sneha	1992-09-18	33	
Vikram	1998-12-25	27	
Anjali	1996-05-14	30	
Karthik	1991-11-30	34	
Meena	1995-08-22	31	
Suresh	1999-02-10	27	
Divya	1996-06-08	30	
+-----+			